

HEBRON, NE, USA

WMO: 722124

Lat: 40.149N

Lon: 97.587W

Elev: 449

StdP: 96.05

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04998

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.4	-14.9	-22.3	0.5	-15.9	-19.8	0.7	-13.3	12.1	-1.2	11.0	-0.1	2.8	340	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	36.0	23.0	34.1	22.8	32.5	22.4	25.4	31.9	24.6	30.9	23.8	29.9	5.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.5	19.4	29.2	22.6	18.3	28.2	22.1	17.7	27.5	80.6	32.0	76.8	31.0	73.3	29.9	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.4	8.4	DB	-22.3	38.7	2.8	1.7	-24.3	39.9	-25.9	40.9	-27.4	41.9	-29.4	43.1	(4)
(5)			WB	-22.6	26.8	2.7	0.9	-24.5	27.4	-26.1	28.0	-27.6	28.4	-29.5	29.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	-2.3	-1.1	6.2	11.5	17.6	23.6	25.4	23.6	20.0	12.5	5.5	-1.3	(6)	
	DBStd	11.14	6.26	6.66	6.25	5.34	4.90	3.55	3.31	3.21	4.59	5.26	5.47	5.90	(7)	
	HDD10.0	1432	382	313	149	46	4	0	0	0	1	33	154	349	(8)	
	HDD18.3	3074	641	543	376	212	76	3	1	2	35	192	386	607	(9)	
	CDD10.0	2100	0	3	33	89	238	408	476	423	299	111	18	1	(10)	
	CDD18.3	701	0	0	1	6	52	161	219	167	84	11	0	0	(11)	
	CDH23.3	7470	0	1	28	101	584	1726	2421	1604	861	138	6	0	(12)	
	CDH26.7	3198	0	0	4	23	214	766	1158	668	331	33	1	0	(13)	
(14) Wind	WSAvg	3.6	3.5	3.8	4.3	4.6	3.9	3.6	2.8	2.6	3.3	3.4	3.6	3.3	(14)	
(15) Precipitation	PrecAvg	710	17	20	51	63	106	102	94	80	77	48	32	19	(15)	
	PrecMax	1150	49	66	204	169	234	271	380	257	239	176	119	62	(16)	
	PrecMin	350	0	0	0	2	21	28	2	6	6	0	0	0	(17)	
	PrecStd	162	13	16	43	35	60	57	71	48	53	42	28	15	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.3	20.9	26.3	29.9	34.7	37.3	37.8	37.1	35.5	31.0	23.8	16.0	(19)	
		MCWB	6.9	9.9	14.6	16.6	18.8	21.6	23.1	23.3	21.3	19.6	13.5	8.7	(20)	
	2%	DB	12.2	16.1	23.0	26.2	31.3	35.0	36.1	34.3	32.6	27.0	20.1	12.6	(21)	
		MCWB	6.0	8.6	13.2	15.8	19.1	22.7	23.7	23.2	21.4	17.1	12.1	7.3	(22)	
	5%	DB	9.0	12.3	20.1	23.6	29.0	33.0	34.2	32.4	30.7	24.7	17.3	10.0	(23)	
		MCWB	4.5	6.6	12.4	14.6	18.7	22.1	23.6	23.2	20.9	15.8	10.6	5.7	(24)	
	10%	DB	6.2	8.4	16.9	21.1	27.0	31.2	32.5	30.3	28.0	22.1	14.7	7.3	(25)	
		MCWB	2.8	4.4	10.8	13.5	17.8	21.5	23.3	22.6	19.5	14.5	9.3	3.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.6	12.5	17.8	18.7	22.9	25.7	26.9	26.4	23.9	21.5	15.5	11.4	(27)	
		MCDB	14.7	16.4	23.3	24.3	29.4	32.5	33.6	32.3	30.4	27.3	18.7	13.9	(28)	
	2%	WB	6.6	9.2	15.8	17.4	21.4	24.4	25.7	25.1	23.0	19.8	13.5	7.7	(29)	
		MCDB	11.5	15.5	19.9	23.7	27.4	31.1	32.2	31.3	29.6	25.0	18.7	11.4	(30)	
	5%	WB	4.7	6.8	13.5	16.0	20.3	23.5	24.9	24.1	22.1	17.2	11.4	5.7	(31)	
		MCDB	8.7	11.6	18.3	21.7	26.4	30.3	31.3	30.0	28.1	22.0	16.2	9.9	(32)	
	10%	WB	3.1	4.7	11.0	14.5	19.0	22.6	24.0	23.4	21.1	15.4	9.6	4.0	(33)	
		MCDB	6.1	8.8	17.2	19.8	25.0	29.2	30.3	28.9	26.2	20.2	14.5	7.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.9	11.2	12.9	13.1	12.3	12.2	11.9	11.8	13.4	13.5	12.8	10.7	(35)	
		MCDBR	15.6	17.7	18.0	16.9	15.8	14.3	13.9	14.0	14.6	17.0	16.6	15.3	(36)	
	5% WB	MCWBR	10.0	10.8	9.6	8.6	6.1	5.2	5.0	5.5	5.4	8.3	9.3	10.2	(37)	
		MCWBR	14.3	16.2	14.7	14.5	13.1	12.4	12.1	12.0	12.2	13.2	15.1	14.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.281	0.289	0.320	0.359	0.386	0.395	0.414	0.413	0.374	0.332	0.287	0.276	(40)	
		taud	2.482	2.470	2.421	2.338	2.336	2.360	2.317	2.318	2.376	2.477	2.525	2.494	(41)	
	Ebn at Noon		895	937	936	914	892	881	860	851	865	872	881	872	(42)	
		Edh at Noon	76	89	104	122	125	122	127	123	109	88	72	70	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.25	3.07	4.11	5.17	6.00	6.80	6.70	5.89	4.81	3.47	2.50	1.91	(44)	
	RadStd		0.20	0.24	0.32	0.30	0.39	0.37	0.39	0.31	0.30	0.35	0.18	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page